ARTWORK PROJECT

Application Flow and File Guide

A concise explanation of how the PHP/MySQL gallery works, which files remain, what was modified, and the responsibility of every project file.

Project	Web Task 04C - Art Gallery
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Stack	PHP, MySQL, HTML, CSS, XAMPP
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Current outcome

The working application and README.md are retained. The earlier Word report, verification document, report screenshots, and document-generation workspace were removed. This PDF is the only new explanatory document.

1. What Was Kept

The application source, database imports, original artwork images, protective configuration, optional test scripts, and README.md remain in place. No PHP, CSS, SQL, image, or README content was changed while preparing this guide.

Kept area	Reason
README.md	Primary project overview, setup instructions, accounts, and verification commands.
PHP application files	Required runtime behavior: login, gallery, add, delete, logout, redirects.
includes/	Shared database, page layout, session, CSRF, and authentication helpers.
assets/ and images/	Presentation styling, fallback image, and 319 Lab 14 artwork images.
sql/	Original art database plus the four required assignment accounts.
scripts/	Optional local setup and automated verification; not required for normal browsing.
.htaccess files	Disable directory listing and block direct access to sensitive source/data files.

2. What Was Removed or Modified

Action	Item	Effect
Removed	Previous .docx report	The earlier Word assignment report is no longer present.
Removed	VERIFICATION.md	Separate verification documentation was removed; README.md remains.
Removed	.report-work/	Deleted screenshots, rendered pages, temporary HTML, cookies, and report builders.
Removed	.docx-venv/	Deleted the temporary Python environment used only for document creation.
Added	output/pdf/artwork-file-flow-guide.pdf	This requested explanation of flow, retained files, modifications, and file roles.
Unchanged	Application source and README.md	No runtime behavior or setup instructions were edited in this request.

Earlier project implementation changes (relative to a basic Lab 14 gallery) include protected customer authentication, the required four accounts, gallery filtering, Bonus image upload, safe deletion, CSRF protection, prepared statements, and optional automated verification.

3. End-to-End Application Flow

1. Entry

index.php or artwork.php loads art-main.php. Because the gallery is protected, an anonymous visitor is redirected to art-login.php.

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2. Sign in

art-login.php displays the shared login form. authenticate.php validates the CSRF token and checks CustomerLogon plus Customers.

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3. Session

On success, includes/auth.php regenerates the session ID and stores the customer identity. The browser is redirected to art-main.php.

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4. Gallery

art-main.php joins Paintings and Artists, applies optional artist/title/year filters, and shows image, title, artist, year, and Delete controls.

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5A. Add

art-add.php validates text, year, MIME type, file size, and CSRF token. It reuses or creates an artist, stores a unique image, and inserts Paintings in a transaction.

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5B. Delete

art-delete.php shows confirmation on GET. A confirmed POST validates CSRF, deletes the selected row, and removes only images created by the add feature.

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6. Sign out

logout.php accepts POST only, validates CSRF, clears the session and cookie, and returns to art-login.php.

4. Data Flow

The shared art_db() helper in includes/app.php reads config.php and opens one mysqli connection. The application uses the existing art database rather than creating a separate gallery database.

Operation	Tables/files involved
Authenticate	CustomerLogon joined to Customers; password_verify checks the stored hash.
List/filter gallery	Paintings joined to Artists; optional WHERE conditions are bound parameters.
Add painting	Artists lookup/insert, Paintings insert, and a new file under images/art/works/square-medium/.
Delete painting	Paintings delete; only an added-* image is eligible for file removal.
Display images	Original or uploaded image; assets/image-unavailable.svg is the browser fallback.

5. Runtime PHP Files

File	Responsibility
index.php	Default entry point; includes art-main.php.
artwork.php	Assignment-named alternate entry point; also includes art-main.php.
art-login.php	GET/HEAD login page; redirects already authenticated customers.
authenticate.php	POST login handler; validates CSRF and credentials, starts authenticated session.
art-main.php	Protected gallery list, artist/title/year filters, ADD link, and Delete buttons.
art-add.php	Protected Bonus form and handler for validated image upload and database insertion.
art-delete.php	Protected confirmation and POST deletion handler with safe upload cleanup.
art-detail.php	Compatibility redirect from older detail links to art-main.php.
logout.php	POST-only sign-out; clears session data and expires the cookie.
config.php	Reads ART_DB_* environment variables and supplies default XAMPP database settings.

6. Shared Include Files

File	Responsibility
includes/app.php	Database connection, input helpers, HTML escaping, CSRF/session functions, page header/footer, flash messages, artwork query, and image path handling.
includes/auth.php	Credential query, password verification, sign-in state, required-login redirect, and login-page composition.
includes/login-form.php	Reusable HTML form fields for user ID, password, CSRF token, and login errors.
includes/.htaccess	Blocks direct browser access to all include files.

7. Presentation and Image Files

File/directory	Responsibility
assets/gallery.css	Simple readable page layout, table styling, filters, buttons, notices, errors, and form controls.
assets/image-unavailable.svg	Fallback graphic when a painting image cannot be loaded.
images/art/works/square-medium/	Stores the original 319 artwork JPEGs and any added-* uploads.

8. Database Files

File	Responsibility
sql/art-small.sql	Complete Lab 14 art database export used to create tables and load artwork data.
sql/part4-users.sql	Idempotently adds or restores user1/pass1 through user4/pass4 with firstN/lastN customer names and password hashes.
sql/.htaccess	Prevents SQL files from being downloaded through Apache.

9. Root Protection and Documentation

File	Responsibility
.htaccess	Turns off directory indexes and blocks config.php plus hidden, SQL, Markdown, Python, shell, and ZIP files.
README.md	Retained project documentation: completed scope, page descriptions, installation, accounts, upload rules, and optional verification commands.

10. Optional scripts/ Files

The scripts directory is useful for setup and testing but is not loaded during ordinary website requests. The site can run without it after the database has been imported.

File	Responsibility
scripts/local-tools.php	CLI-only helpers for XAMPP paths, MySQL socket access, child processes, and consistent error handling.
scripts/setup-local.php	Imports art-small.sql when needed, reapplies part4-users.sql, and prints painting/artist totals.
scripts/create-login-sql.php	Regenerates part4-users.sql with fresh password_hash values for pass1 through pass4.
scripts/http-checks.php	Reusable cURL test client, assertions, cookie handling, login helper, and test runner.
scripts/gallery-checks.php	Gallery-specific test helpers for parsing painting rows, fixtures, uploads, and redirects.
scripts/verify.php	Five gallery test groups: entry pages, filters, add/delete persistence, validation, and protected files.
scripts/verify-auth.php	Six authentication test groups: four users, invalid credentials, HTTP methods, protected routes, token isolation, and logout.
scripts/.htaccess	Blocks browser access to all test/setup scripts.

11. Which Files Are Required to Run?

Category	Keep for runtime?	Notes
PHP files and includes/	Yes	Core application behavior.
config.php	Yes	Database connection settings.
assets/ and images/	Yes	CSS and artwork media.
MySQL database	Yes	Must be imported before the site can query data.
sql/	Recommended	Needed to recreate or submit the database; not queried at runtime.
README.md	Recommended	Setup and handoff information.
scripts/	Optional	Only setup and automated verification; not required by page requests.

12. Quick Run Summary

- Start Apache and MySQL in XAMPP.
- Import sql/art-small.sql and then sql/part4-users.sql if the art database is not already installed.
- Open http://localhost/artwork/art-login.php.
- Sign in with user1/pass1 (or user2-pass2 through user4-pass4).
- Use the gallery to filter, add, and delete paintings, then sign out.

README.md remains the authoritative setup note. The scripts folder is optional. The PHP application, database files, CSS, and artwork images preserve the working solution and assignment submission.